Skill, practice and what else: fits with accuracy

16 contrasts; 14 with accuracy and error RTs. Extended Wong-Wang model with evidence, choices and the E/I control signal of Lo, Wang & Wang (2015).

Question	Measured	
Skill + practice produce correct RTs, accuracy and error RTs adequately (p > .05)	5 of 16 contrasts	a few
Skill + practice + E/I control signal together adequate (p > .05)	8 of 16 contrasts	a few
Where skill + practice fail: lowest-AIC adequate single addition	drive: 3; ndt: 1; none adequate: 7	
Where skill + practice fail: adequate with each addition	drive: 4; evidence: 0; control: 3; ndt: 4; control + ndt: 3; drive + ndt: 4	of 11

Per contrast

Id	Task	Change	Accuracy A → B	Skill + practice	SP + E/I (p)	Lowest AIC
P1	brightness discrimination	Time on task	0.712 → 0.703	no; + drive	no (0.043)	SP + drive + ndt
P2	lexical decision	Time on task	0.910 → 0.866	no; none adequate	no (0.0006)	SP + control + ndt
P3	lexical decision (BLP)	Time on task	not observed	both	yes (0.35)	SP
P4	motion discrimination	Time on task	0.675 → 0.704	no; + ndt	yes (0.32)	SP + ndt
P5	visual search	Time on task	0.977 → 0.977	either	yes (0.82)	SP
P6	visual working memory	Time on task	0.826 → 0.821	both	yes (0.44)	SP + evidence
P7	Stroop and Simon	Time on task	0.980 → 0.971	no; + drive	yes (0.19)	SP + drive
P8	cued task switching	Time on task	0.840 → 0.871	no; none adequate	no (0)	SP + evidence
P9	attention network…	Time on task	0.968 → 0.970	either	yes (0.62)	SP + evidence
A1	brightness discrimination	Instruction	0.699 → 0.722	no; none adequate	no (0)	SP + ndt
A2	lexical decision	Instruction	0.843 → 0.938	no; none adequate	no (0)	SP + control + ndt
F1	lexical decision	Word frequency	0.766 → 0.947	no; none adequate	no (0)	SP + evidence
S1	StarCraft 2 by league	Expertise	not observed	no; + drive	yes (0.08)	SP + drive
S2	motion discrimination	Expertise	0.670 → 0.679	skill	yes (0.98)	SP + ndt
D1	motion discrimination…	Stimulus strength	0.642 → 1.000	no; none adequate	no (0)	SP
D2	motion discrimination	Stimulus strength	0.728 → 0.976	no; none adequate	no (0)	SP + evidence

1. Skill = drive spread u0_sd (u0 fixed); practice = stored instances logN; common drive = u0 and u0_sd scaled together; evidence = difference between the pools' drive (Hz); control signal = balanced synaptic input, strength S and ratio R, from the f-I table derived from the paper's neuron; non-decision time = ter_mu.

2. Each contrast's A is calibrated to its pooled correct RTs and accuracy, with evidence and the instance route's noise free. Chi-square over correct-RT percentiles, accuracy change and error-RT percentiles (where A and B each have >= 100 errors), with data and simulation error; AIC = chi-square + 2 x changes.

3. No accuracy: P3 (BLP errors have no RT in DataPrep) and S1 (StarCraft 2); these are fitted on correct RTs only.

4. Optimiser checks pending: D1 (practice at its bound; adding evidence predicts accuracy falling) and P8 (E/I fit at strength 0 scores far better than skill + practice, which it contains); their fits may not be the best attainable.

5. Sources: 4_output/acc_mixture_summary.csv, acc_mixture_fits.csv, pooled_accuracy_profiles.csv.

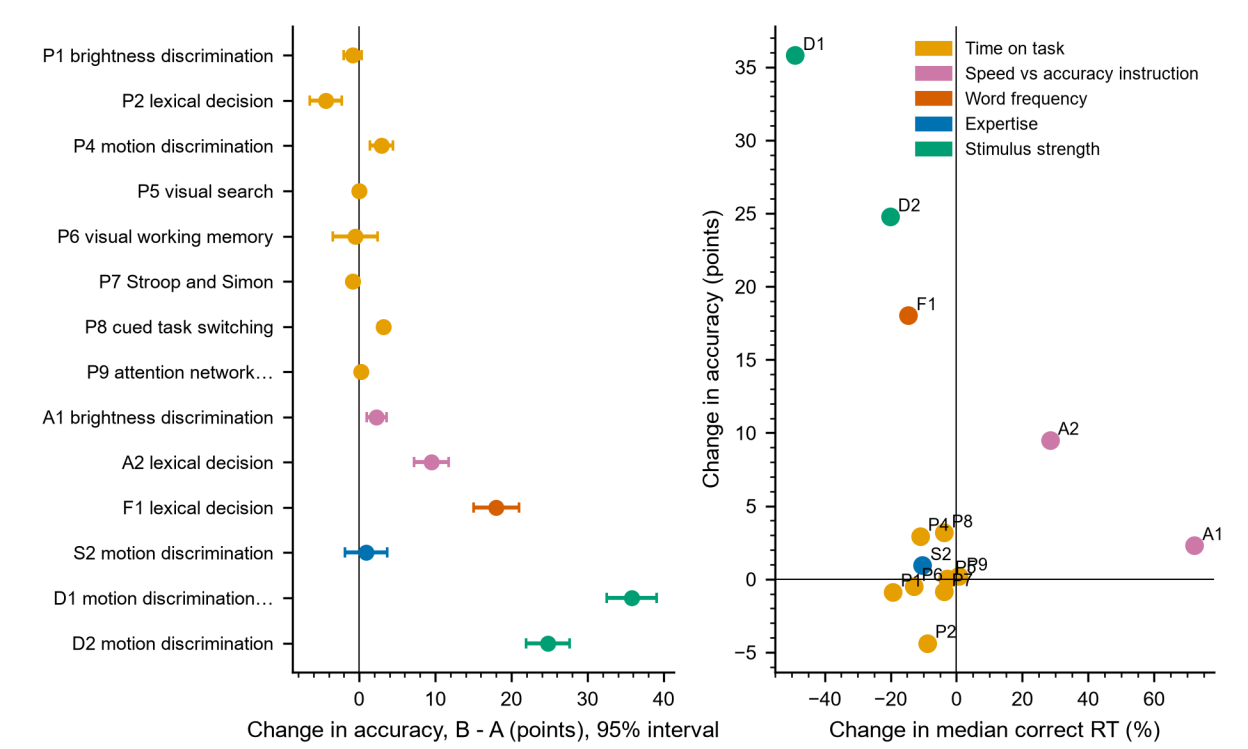


Figure 1. The data. Left: change in accuracy from A to B, pooled over units, with 95% cluster-bootstrap intervals. Right: change in accuracy against change in median correct RT. Upper left = faster and more accurate; lower left = faster and less accurate; upper right = slower and more accurate.

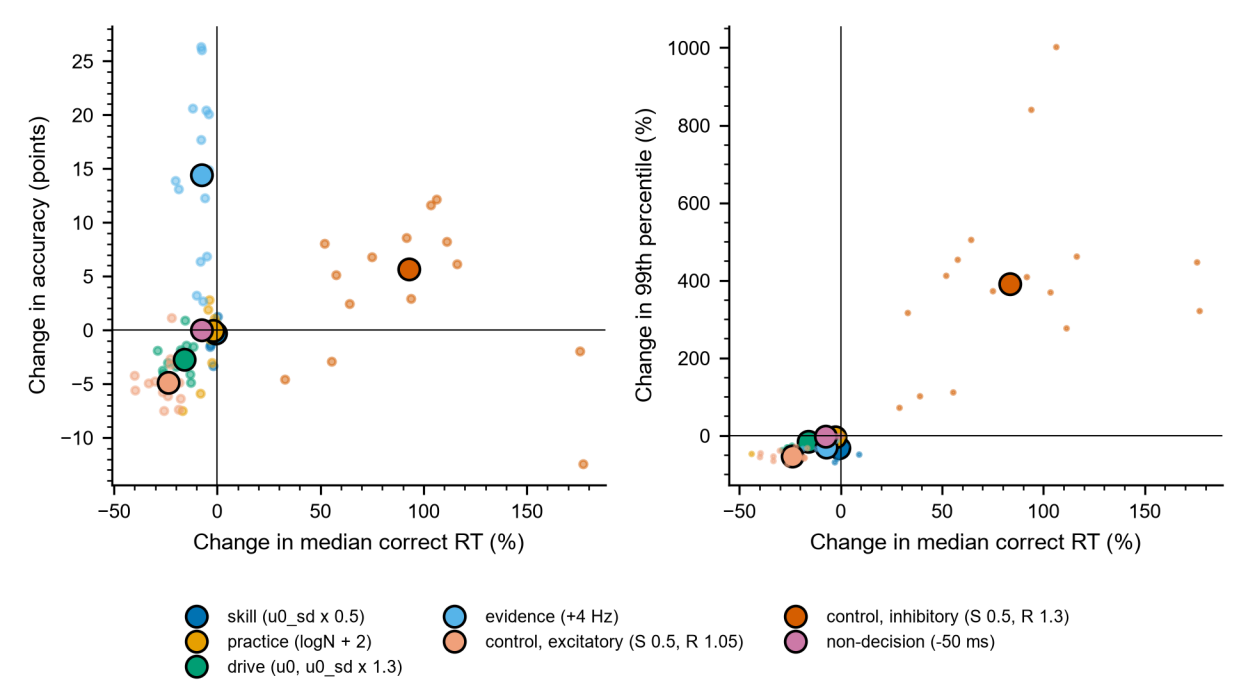


Figure 2. What each ingredient does. One step of each change from every contrast's calibrated A (small points; large = median over contrasts). Left: speed against accuracy, where ingredients that make the same RT change separate by their effect on accuracy. Right: the median against the 99th percentile.

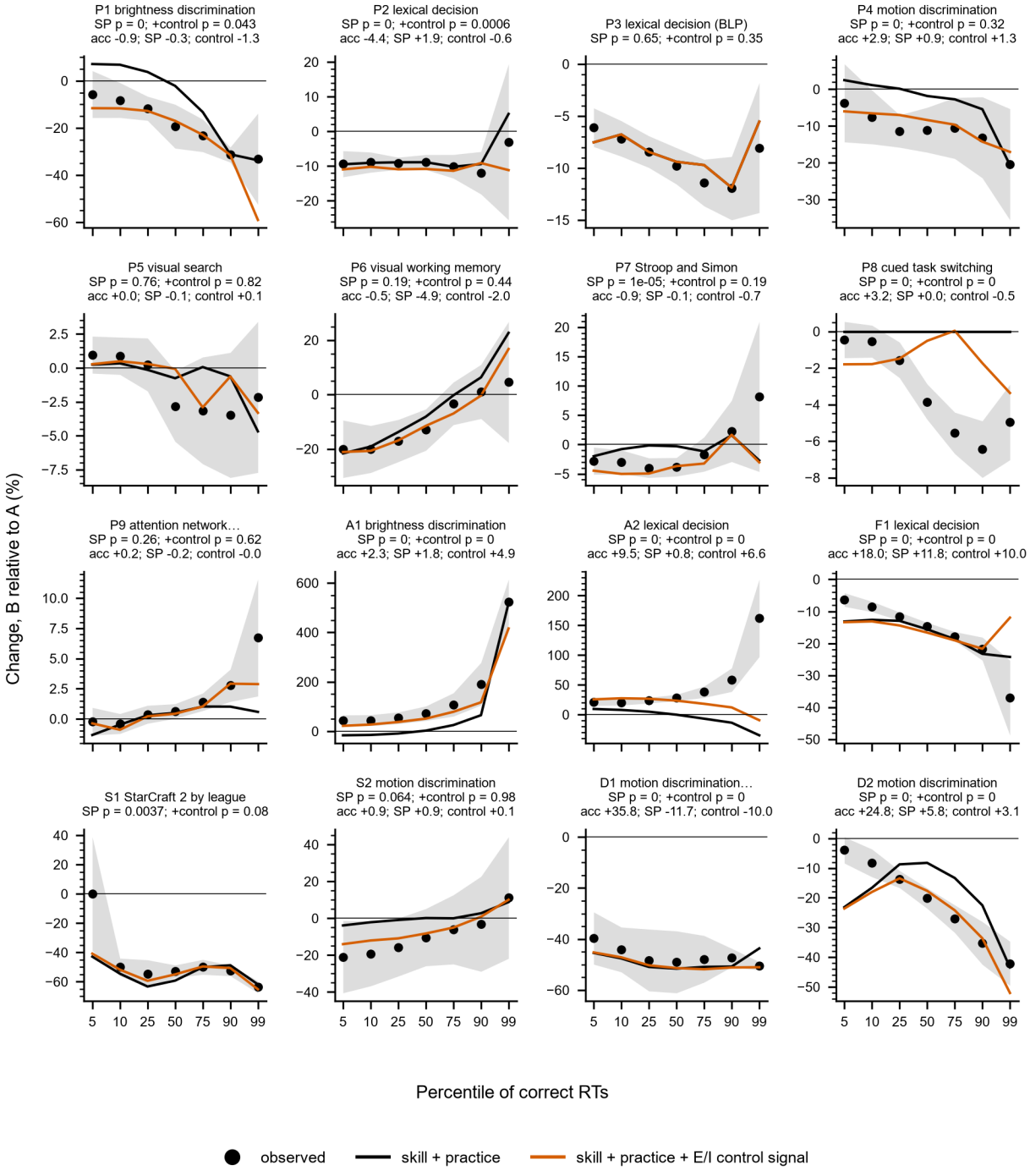


Figure 3. The fits. Observed change in correct-RT percentiles (points; band = 95% interval), skill + practice (black) and skill + practice + E/I control signal (orange), each fitted to correct RTs, accuracy and error RTs together. Titles: chi-square p of each, and the observed and predicted change in accuracy (points).

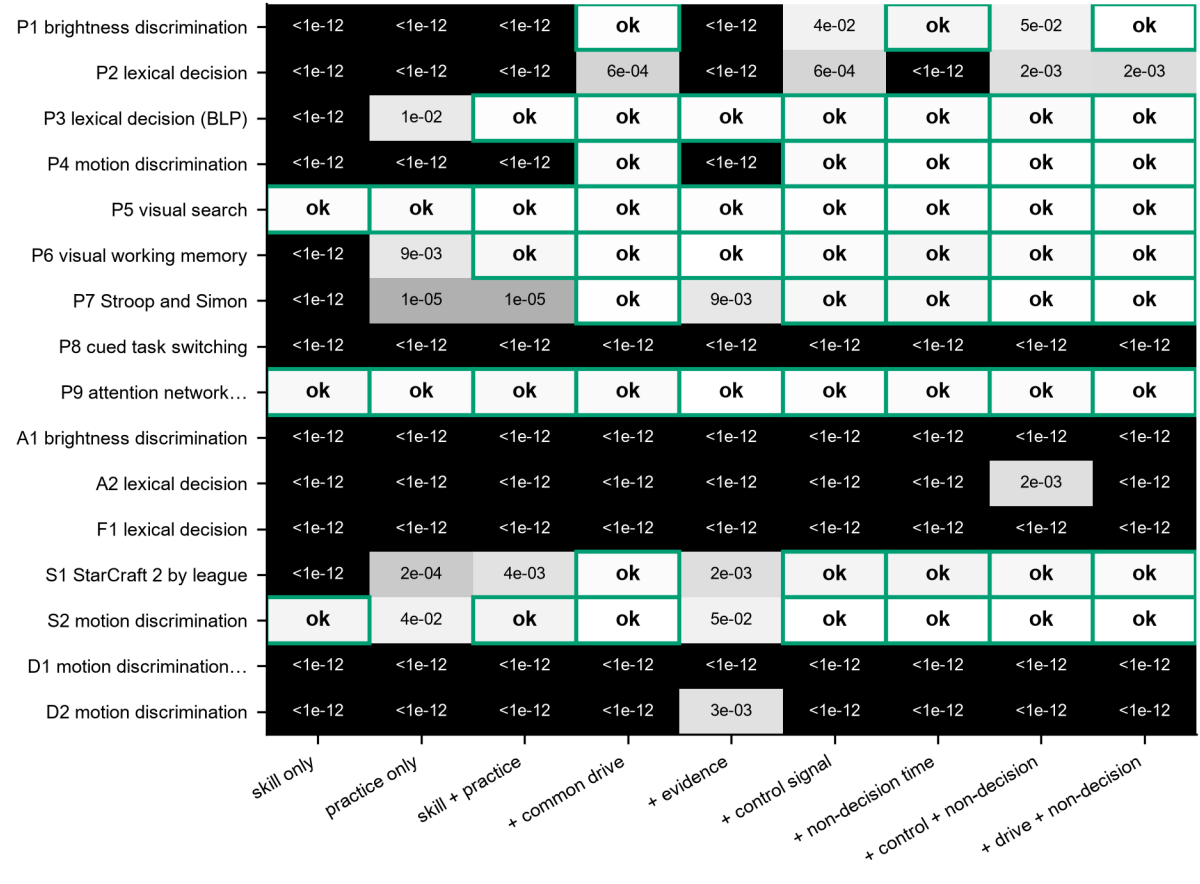


Figure 4. Which models are adequate (ok in a green box: chi-square p > .05). Shading: -log10 p, capped at 12. Contrasts with accuracy are judged on correct RTs, accuracy and error RTs together.

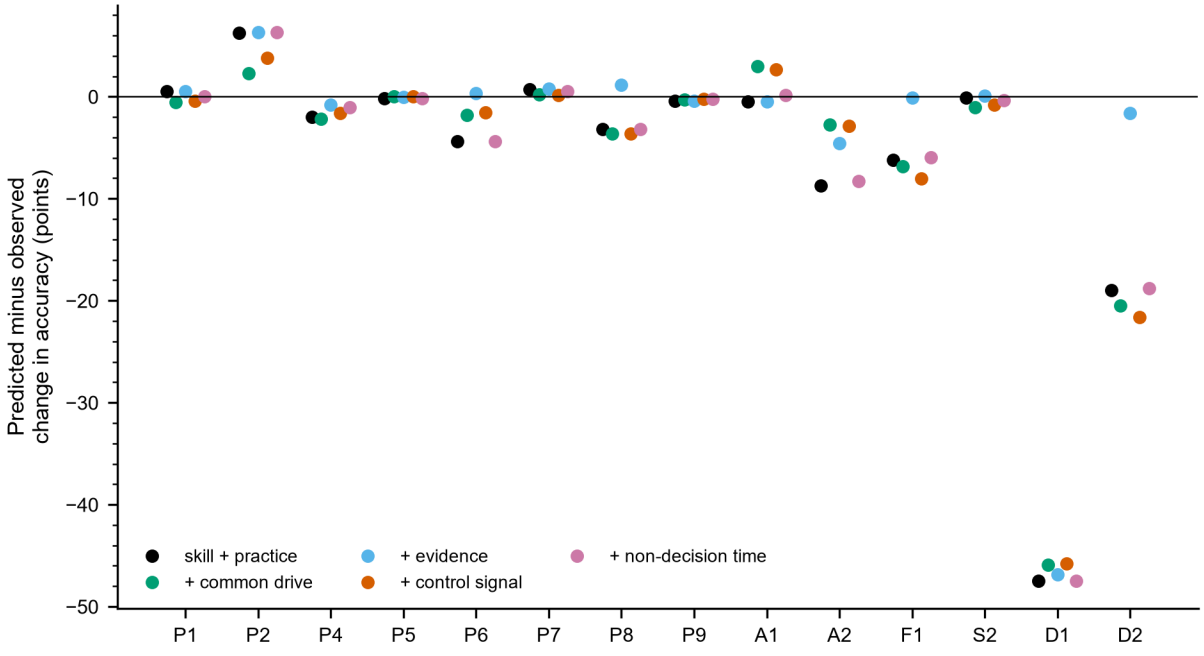


Figure 5. How well each fit reproduces the change in accuracy: predicted minus observed, in percentage points, for skill + practice and each single addition.

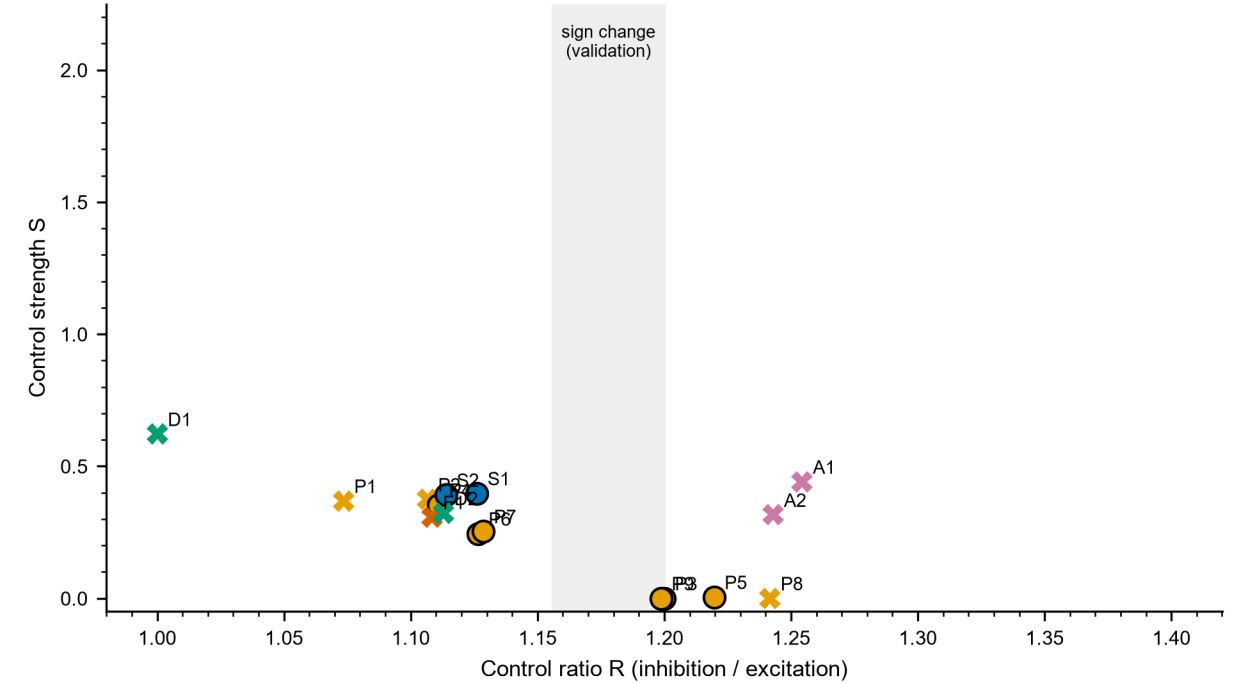


Figure 6. The control signal in the skill + practice + control fit of each contrast. Left of the grey band the signal speeds decisions (net excitation); right of it, slows them (net inhibition) (process/output/check_06.txt). Circle = adequate fit; cross = not. Colours: kind of contrast, as in Figure 1.

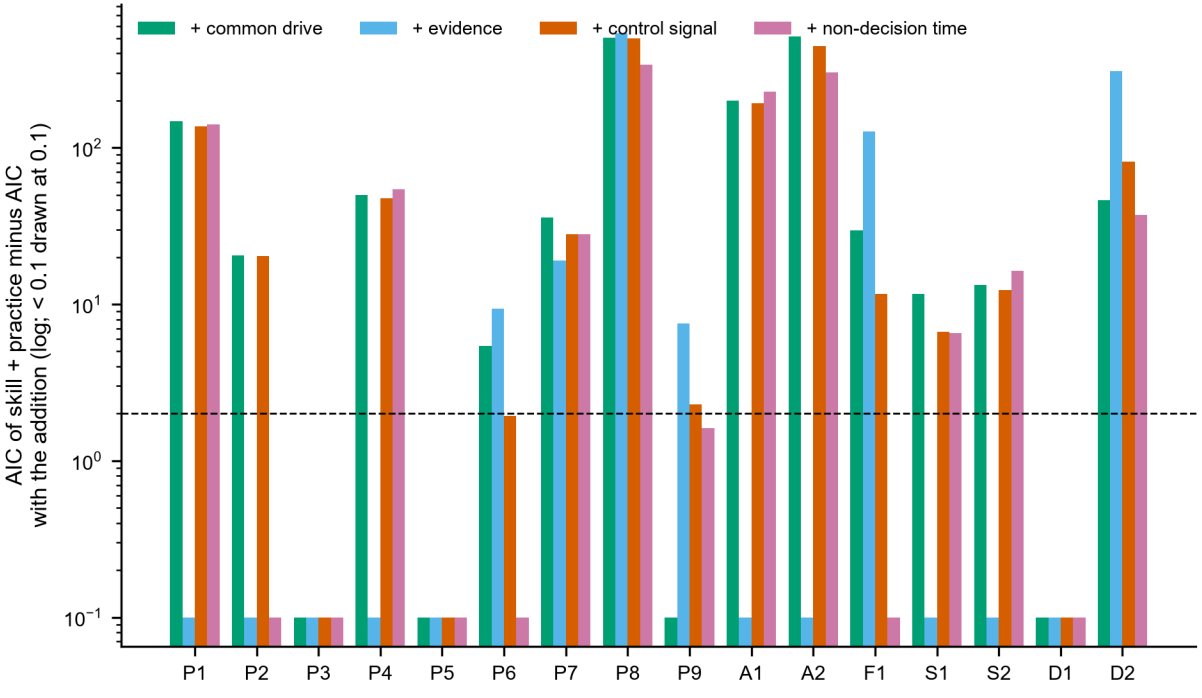


Figure 7. How much each single addition improves on skill + practice (drop in AIC; dashed line = 2).